

Multilinear Algebra and Category Theory: Summary Notes

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1 Categorical foundations

1.1 Objects and morphisms

Definition 1.1 (Category). A *category* $\mathcal{C}$ consists of the following data:

- A class of *objects*, denoted $\text{Ob}(\mathcal{C})$, for example sets, groups, rings.
- For every $A, B \in \text{Ob}(\mathcal{C})$, a set of *morphisms* or *arrows*, denoted $\text{Mor}_{\mathcal{C}}(A, B)$ or $\text{Hom}_{\mathcal{C}}(A, B)$,
- For every $A, B, C \in \text{Ob}(\mathcal{C})$, a *composition* operation

$$\begin{aligned}\circ : \text{Hom}_{\mathcal{C}}(A, B) \times \text{Hom}_{\mathcal{C}}(B, C) &\rightarrow \text{Hom}_{\mathcal{C}}(A, C) \\ (f, g) &\mapsto gf\end{aligned}$$

such that

- Composition is associative (when defined), and
- For every $A \in \text{Ob} \mathcal{C}$ there is an identity morphism $1_A \in \text{Hom}_{\mathcal{C}}(A, A)$ such that $1_A f = f$ and $g 1_A = g$ for every $f \in \text{Hom}_{\mathcal{C}}(B, A)$ and $g \in \text{Hom}_{\mathcal{C}}(A, B)$

Key Idea: Purpose and intuition of categories

An object is an abstract structure, for example a set, group or ring. A morphism is a structure-preserving map, for example a function or a group/ring homomorphism.

Example: Categories

- **Sets**: sets and functions, with function composition. Any category consisting of sets and functions (**Groups**, **Rings**, **Top**, etc) will have function composition as the composition rule.
- **Groups**: groups and group homomorphisms.
- **AbGrps**: abelian groups and group homomorphisms.
- **Rings**: rings and ring homomorphisms.
- **Top**: topological spaces and continuous maps.
- **Man**: smooth manifolds and diffeomorphisms.

Warning / Pitfall: Morphisms are not always functions.

Consider a category $\mathcal{C}$ with a single object $\star \in \text{Ob}(\mathcal{C})$ whose morphisms are elements g of a group G which compose via group multiplication. The morphisms are not themselves functions, but rather elements of the set G .

Key Idea: Types of morphisms.

Functions may be injective, surjective, and invertible. We extend this to morphisms $f : a \rightarrow b$.

Monomorphism (monic). Left-cancellable, i.e. $f g_1 = f g_2 \implies g_1 = g_2$. *Extends injectivity.*

Epimorphism (epic). Right-cancellable, i.e. $g_1 f = g_2 f \implies g_1 = g_2$. *Extends surjectivity.*

Bimorphism (mono-epi). Both monic and epic. *Extends bijectivity.*

Section. Left-invertible, i.e. there is $g : b \rightarrow a$ such that $g f = 1_a$. Necessarily a monomorphism.

Retraction. Right-invertible, i.e. there is $g : b \rightarrow a$ such that $f g = 1_b$. Necessarily an epimorphism.

Isomorphism. Both a section (left-invertible) and retraction (right-invertible) by the same inverse. *Extends invertibility.*

Proof Strategy: Relationships between types of morphisms.

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Warning / Pitfall: Bijective morphisms are not always isomorphisms.

Certainly if a morphism is not a function, it does not make sense to ask if it is a bijection. But even still, there are functions which are bijective morphisms but their inverse is not a morphism. For example in the category of topological spaces, the function $f : [0, 2\pi) \rightarrow S^1$ with $f(\theta) = (\cos \theta, \sin \theta)$ is continuous and bijective, but not a homeomorphism, because the inverse is not continuous.

1.2 Functors and natural transformations

2 Algebraic foundations

3 Multilinear algebra

4 Bilinear forms

5 Classification and structure theorems

6 Reference: universal properties glossary
